1 GAME OBJECTIVES / BREIF

1.1 GAME BRIEF:

Hues ‘n Dash is a pixel-style 2D puzzle-platformer where players navigate color-coded levels under time pressure while being pursued by enemies.

The core mechanic revolves around collecting a coloured object that allows the player to pass through barriers of the same color. Since the player can carry only one object at a time, strategic decision-making and quick movement are essential to complete each level.

The game blends arcade-style action with puzzle-solving elements to create a fast-paced yet thoughtful gameplay experience.

1.2 NARRATIVE BRIEF

Hues ‘n Dash is set inside a corrupted digital system where the protagonist, programmer, must restore order by solving logic-based challenges. Each level represents a section of broken or corrupted code. To progress, the programmer must collect specific color-coded objects that function like logic keys, allowing them to pass through matching color barriers. These barriers represent logic gates within the system, reinforcing the programming metaphor.

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1.3 GAME OBJECTIVES:

- Collect the correct color-coded object required to pass through matching barriers.
- Strategically manage the single-item carrying constraint to solve puzzles.
- Avoid being caught by the pursuing Bug (enemy).
- Complete each level within the given time limit.
- Reach the designated finish point to restore stability to that section of the system.

2 GAMEPLAY

2.1 OVERVIEW:

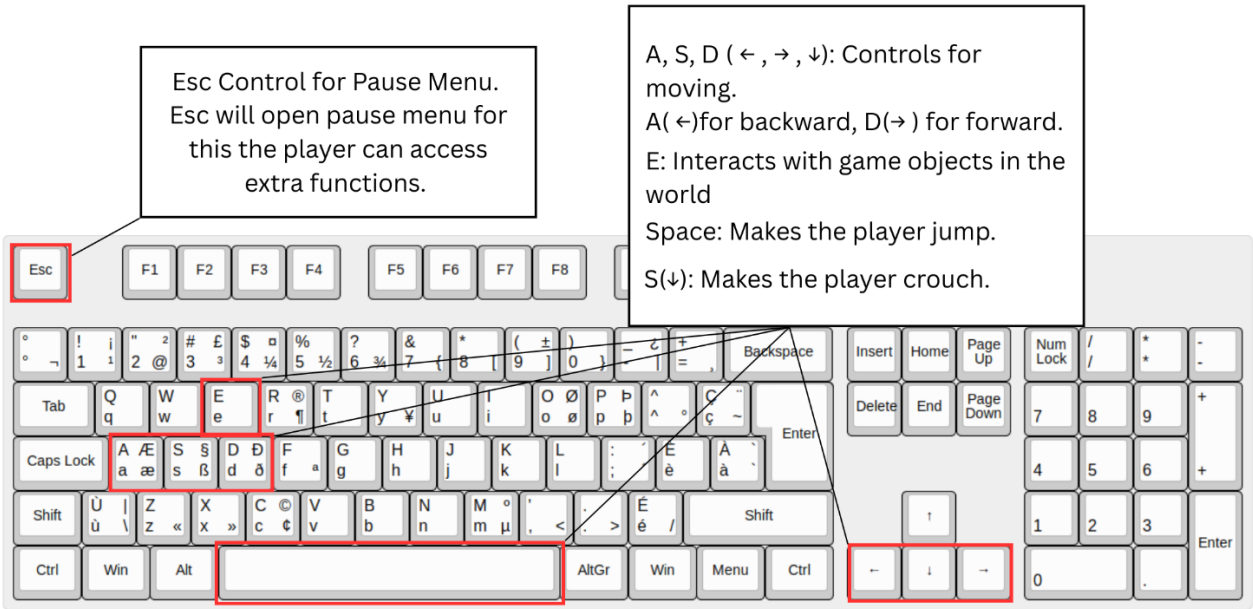
Gameplay will be designed for PC use, utilizing keyboard controls for character movement and interaction. The control scheme will follow a familiar 2D platformer layout, similar to classic retro PC games, making it intuitive and easy to learn for players accustomed to standard keyboard controls.

The player will use directional keys for movement and a dedicated key for jumping and interacting with objects. The mechanics will focus on smooth horizontal movement, responsive jumping, and precise object pickup actions.

Character movement and interactions will be supported by pixel-based animations to ensure visual feedback and clarity during gameplay. Variables such as movement speed, jump height, gravity, and enemy chase speed will be carefully adjusted to maintain balanced difficulty and responsive control.

The overall aim is to create a control system that feels simple, responsive, and consistent with retro arcade-style platformers while supporting the game's logic-based puzzle mechanics.

2.2 LAYOUT



3 GRAPHICS/ ART STYLE/SOUND DESIGN

3.1 GRAPHICS

Hues 'n Dash will feature a 2D pixel-art visual style. The game environment will be designed using simple but vibrant color palettes to clearly represent the color-coded mechanics. Each barrier and object will have distinct colors to ensure visual clarity and immediate player recognition.

The interface will be minimal, displaying only essential information such as the timer and level indicators. Visual feedback effects will be used when the player picks up an object, unlocks a barrier, or gets caught by the enemy.

The overall goal is to maintain readability and strong contrast while preserving a nostalgic arcade aesthetic.

3.2 ART STYLE

The art style will be inspired by classic 8-bit and 16-bit platformer games. Characters, enemies, and environmental elements will be created using pixel-based sprites with simple animations.

The main character design draws inspiration from a programmer theme, featuring a clean and recognizable pixel silhouette. The enemy ("Bug") will have a slightly contrasting design to visually communicate danger.

Backgrounds will reflect a digital world setting, potentially including grid patterns, circuit-like elements, neon accents, and subtle glitch effects to reinforce the corrupted system theme.

The artistic direction focuses on simplicity, clarity, and strong color identity.

3.3 SOUND DESIGN

The sound design will complement the retro arcade atmosphere. Background music will be simple, loop-based, and energetic, inspired by classic pixel-era games.

Sound effects will include:

- Object pickup sound
- Enemy alert or chase cue
- Level completion sound
- Failure / restart sound

Audio feedback will be clear and responsive to enhance player engagement and reinforce gameplay actions. Royalty-free audio assets will be used and optimized for performance.

4 CHARACTERS

The enemy (Bug)

The Main Character (Programmer)

4.1 CHARACTER MECHANICS

The player controls a programmer character navigating through a corrupted digital environment. The character mechanics are designed to be simple, responsive, and suitable for a retro 2D puzzle-platformer.

4.2 MOVEMENT MECHANICS

- The character can move left and right using keyboard controls.
- The character can jump to navigate platforms and avoid obstacles.
- Movement speed and jump height are predefined and balanced for precision platforming.
- Gravity and collision detection ensure realistic and consistent movement behavior.

4.3 OBJECT INTERACTION MECHANICS

- The character can pick up color-coded objects placed within the level.
- The player can carry **only one object at a time**.
- Picking up a new object automatically replaces the currently held object.
- While carrying an object, the character can pass through barriers that match the object's color.
- If the object color does not match the barrier, passage is blocked.

4.4 Enemy Interaction

- The “Bug” enemy actively chases the player within the level.
- If the enemy collides with the player, the level restarts.
- The player must use movement and logical planning to avoid the enemy while solving puzzles.

4.5 Time Constraint

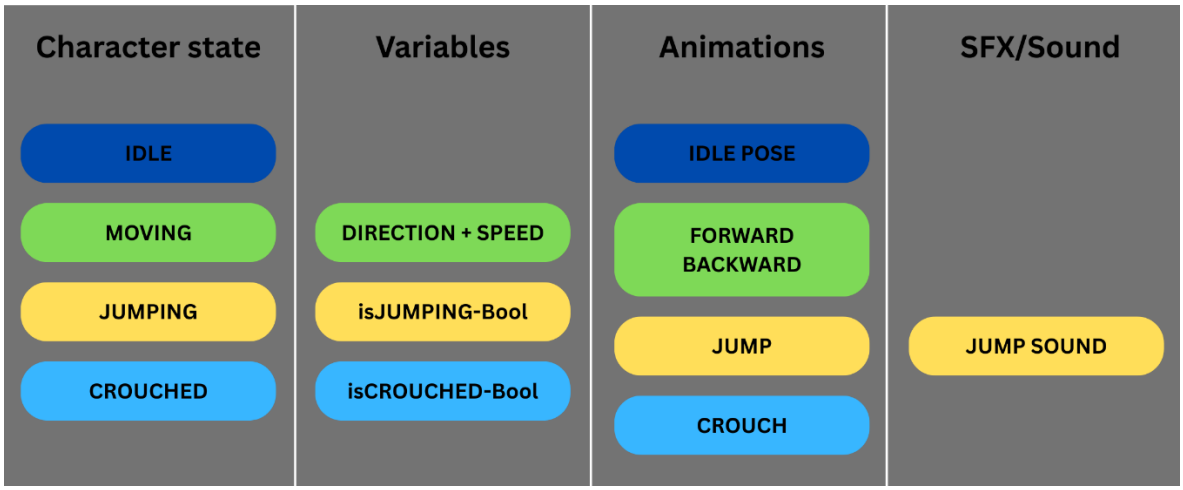
- Each level includes a countdown timer.
- If the timer reaches zero, the level resets.
- This mechanic increases urgency and encourages efficient decision-making.

4.6 Level Completion

- The character must reach the designated finish point after solving the required logic barriers.
- The level is considered complete only if all necessary progression conditions are met.

4.7 Character Animation

The character animation in Hues ‘n Dash will follow a pixel-art style, designed to be simple, readable, and responsive. Animations will provide clear visual feedback to the player while maintaining the classic arcade aesthetic. Below is a graph showing what needs to happen in each character state.



5 PROJECT PLAN

5.1 DEVELOPMENT METHODOLOGY

The project will follow the **Agile development model**, where development is carried out in iterative stages. Each iteration includes planning, design, implementation, testing, and review. This approach allows flexibility, continuous improvement, and manageable progress tracking.

5.2 PROJECT PHASES

Sprint 0 – Planning & Resource Analysis

Identify hardware & software resources

Choose Agile model

Decide game mechanics at high level

Create GitHub repository

Sprint 1 – Core Gameplay

Design player movement

Implement base level

Test

Sprint 2 – Puzzle System

Design color-item logic

Implement door mechanics

Test

Sprint 3 – Enemies & Timer

Design enemy behavior

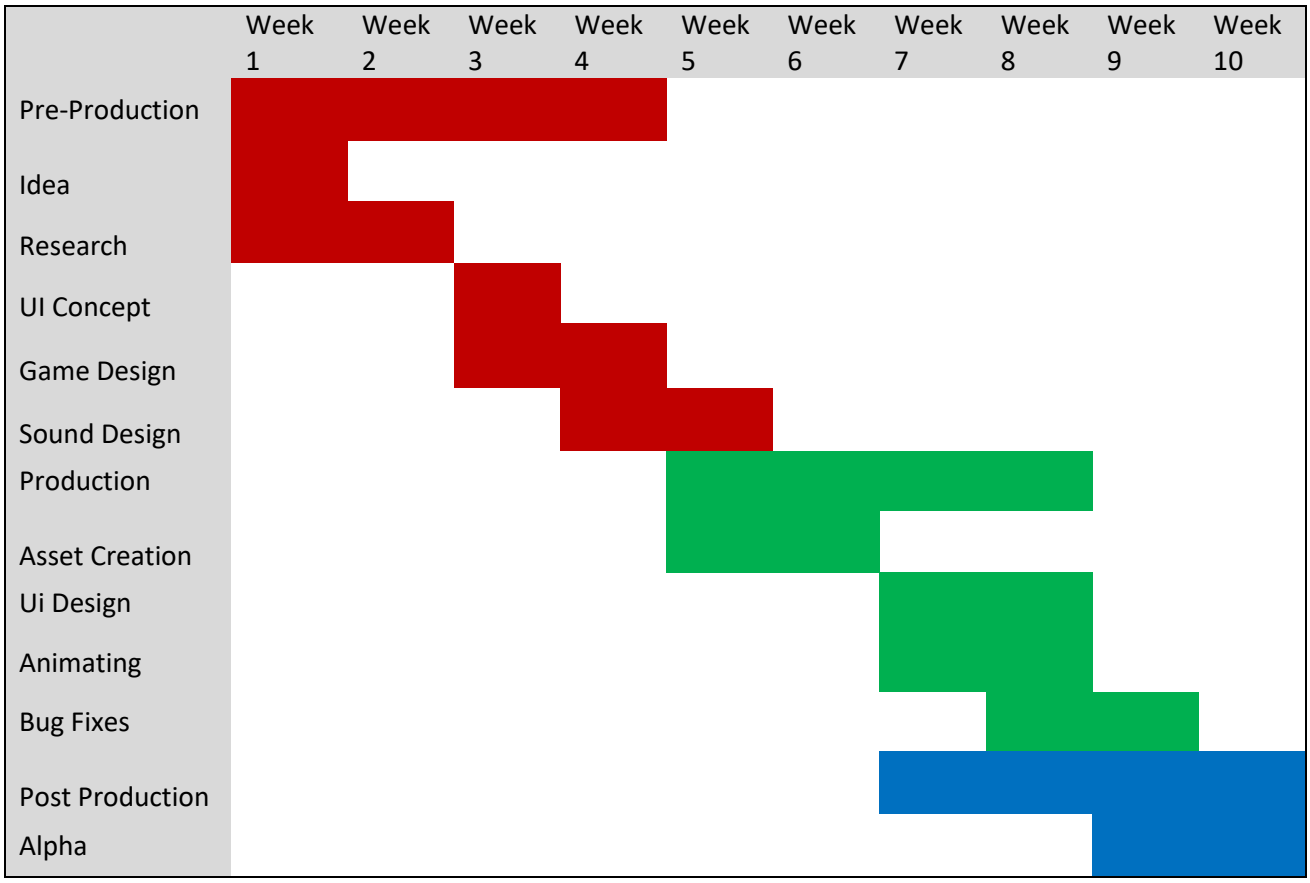
Implement follow logic

Test

Sprint 4 – Polish & Finalization

UI & sound

Bug fixing

Documentation

6 GANNT CHART